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Will power-seeking AGIs harm human society?

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Abstract
Many have argued, based on the Instrumental Convergence Thesis, that Artificial General Intelligences (AGIs) will exhibit power-seeking behavior. Such behavior, they warn, could harm human society and pose existential threats—namely, the risk of human extinction or the permanent collapse of civilization. These arguments often rely on an implicit and underexamined assumption: that AGIs will develop world models—internal representations of world dynamics—that resemble those of humans. We challenge this assumption. We argue that once the anthropomorphic assumption—that AGIs' world models will mirror our own—is rejected, it becomes unclear whether AGIs would pursue the types of power commonly emphasized in the literature, or any familiar types of power at all. This analysis casts doubt on the strength of existing arguments linking the Instrumental Convergence Thesis to existential threats. Moreover, it reveals a deeper layer of uncertainty. AGIs with non-human world models may identify novel or unanticipated types of power that fall outside existing taxonomies, thereby posing underappreciated risks. We further argue that world model alignment—an issue largely overlooked in comparison with value alignment—should be recognized as a core dimension of AI alignment. We conclude by outlining several open questions to inform and guide future research.

Keywords AGI risk · Instrumental convergence thesis · Existential catastrophe · AI alignment · Anthropomorphism · World model alignment

1 Introduction

Artificial Intelligence (AI) systems are beginning to match or even surpass human performance in specialized tasks, raising concerns about Artificial General Intelligences (AGIs) capable of equaling or exceeding human capabilities across a broad range of domains (Fitzgerald et al. 2020; Morris et al. 2023; Mitchell 2024). Researchers have been investigating potential long-term risks posed by AGIs, especially the possibility of existential catastrophe. These concerns have prompted growing calls to develop advanced AI systems that are both safe and aligned (Russell 2019; Hendrycks and Mazeika 2022; McLean et al. 2023; Ji et al. 2024; Shah et al. 2025).

A central concern in the literature is that AGIs—being instrumentally rational and highly capable—will pursue certain subgoals regardless of their specific final objectives. This claim is known as *the Instrumental Convergence Thesis* (Omohundro 2008; Bostrom 2014). In particular, AGIs may pursue various types of power, such as self-preservation, goal preservation, cognitive enhancement, technological development, and resource acquisition (Omohundro 2008; Bostrom 2014; Carlsmith 2023). Building on this idea, several authors argue that AGIs may attempt to avoid shutdown, acquire resources, manipulate humans, or resist modification (Turner et al. 2019; Ngo et al. 2023; Dung 2024; Carlsmith 2023). The pursuit of such power could significantly harm human society, potentially leading to existential threats (Dung 2024; Field 2025; Carlsmith 2023).

Existing arguments warn against anthropomorphizing AGIs' final goals. However, they often implicitly assume that AGIs' world models—their internal representations of world dynamics—will resemble those of humans. This paper critically examines this often-overlooked form of anthropomorphism about AGIs' world models and challenges it. We argue that, once this assumption is rejected, it becomes unclear whether AGIs will pursue the types of power emphasized in the literature, or any familiar types of power at all.

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This analysis questions the strength of existing arguments linking the Instrumental Convergence Thesis to existential threats. It also reveals a deeper layer of uncertainty and highlights an overlooked risk: AGIs with non-human world models may identify novel and potentially dangerous types of power that fall outside current taxonomies. To address this, we argue that world model alignment—largely neglected compared to value alignment—should be recognized as a crucial dimension of AI alignment. We also propose several open questions to advance future research.

To clarify, while this paper uses the term "AGIs" following the convention, the discussion broadly applies to any sufficiently advanced AI system capable of identifying instrumentally useful subgoals. Thus, the insights are pertinent even for those skeptical that fully general AGIs will emerge soon.

This paper proceeds as follows. Section 2 reconstructs the argument linking the Instrumental Convergence Thesis to existential threats. Section 3 challenges the anthropomorphic assumption underlying AGIs' world models. Section 4 shows how rejecting this assumption weakens existing threat arguments. Section 5 introduces new risks posed by non-human world models. Section 6 argues for recognizing world model alignment as a core alignment issue and presents open research questions. Section 7 concludes.

2 Power-seeking AGIs and existential threats

Given that AGIs do not currently exist, any claims about their behavior must rely on assumptions that guide our predictions. The suggestion that AGIs will exhibit power-seeking behavior, and that such behavior could pose a serious threat to human society, is typically based on the assumption that AGIs will be instrumentally rational and possess "instrumental reasoning capacities far exceed those of any human" (Bostrom 2014). The instrumental rationality assumption has been defended by many (Legg and Hutter 2007). Accordingly, this paper proceeds with this assumption. This section reconstructs the arguments linking such AGIs to power-seeking behavior and existential threats.

Instrumental rationality is about selecting means in pursuit of given ends. AGIs' ends—normally referred to as desires or *final goals*—need not resemble those of humans.[1] As many have emphasized, we should *avoid anthropomorphism*: AGIs may have final goals that are substantially different from those of humans. Moreover, it has been suggested that AGIs' level of intelligence is independent of their final goals, a claim known as *the Orthogonality Thesis* (Bostrom 2014). In principle, then, AGIs might possess a wide range of final goals, such as maximizing the number of paperclips, accurately counting sand grains, etc.

Although we do not know what final goals AGIs will have, many suggest that we can still predict certain effective means they are likely to pursue—commonly referred to as *subgoals.* The idea is that since AGIs are instrumentally rational and highly capable, they will recognize certain subgoals as effective means for achieving their final goals. Instrumentally *convergent* subgoals are those that are useful *for achieving a broad range of final goals*. They are, therefore, likely to be pursued by many AGIs regardless of their specific final goals. This idea underlies *the Instrumental Convergence Thesis*, introduced by Bostrom[2]:

Instrumental Convergence Thesis. Several instrumental values can be identified which are convergent in the sense that their attainment would increase the chances of the agent's goal being realized for a wide range of final goals and a wide range of situations, implying that these instrumental values are likely to be pursued by many intelligent agents. (Bostrom 2014, p. 109).

Power has been widely suggested as an instrumentally convergent subgoal, given its general usefulness in achieving a broad range of final goals. Consequently, AGIs are expected to exhibit power-seeking behavior. For instance, Carlsmith (2023, p. 11) writes: "…[P]ower, almost by definition, is extremely useful to accomplishing objectives. Therefore, to the extent that an agent is engaging in unintended behavior in pursuit of problematic objectives, it will generally have incentives, other things equal, to gain and maintain types of power in the process."

AGIs are expected to pursue various types of power, including self-preservation, goal preservation, cognitive enhancement, technological development, and resource acquisition. In light of these power-seeking tendencies, Turner et al. (2019) highlight the incentive to avoid shutdown. Ngo et al. (2023) suggest that AGIs may attempt to acquire money, manipulate humans, and prevent themselves from being modified. Similarly, Carlsmith (2023) lists money, compute, energy, and social influence as powers likely to be pursued. Moreover, Dung (2024) emphasizes that AGIs might want to disempower humanity (also see Cotra 2021; Carlsmith 2023).

These considerations have led many to warn that power-seeking AGIs may pose serious threats to human society. For example, Dung (2024) suggests that humanity could be

[1] For objections to, and defenses of, the claim that AGIs will have desires or final goals, see Goertzel (2015) and Dung (2023). It is worth noting that our discussion does not depend on AGIs actually having desires (or beliefs); rather, it applies as long as AGIs behave in ways consistent with having such states.

[2] Similar ideas were discussed earlier by Omohundro (2008).

permanently disempowered by 2100. Similarly, Carlsmith (2023) estimates that the probability of such an existential catastrophe occurring by 2070 exceeds 0.1. Field (2025) further reports that a large majority of surveyed AI experts (78%) agree or strongly agree that "technical AI researchers should be concerned about catastrophic risks." Moreover, experts familiar with the Instrumental Convergence Thesis were less likely to consider those risks overstated.

The types of power AGIs are expected to pursue play a crucial role in arguments linking the Instrumental Convergence Thesis to existential threats. After all, many living organisms could be described as "power-seeking" in some sense, yet few pose serious danger to human society. The reason is that the types of power they pursue are generally not inherently dangerous. Consider, for instance, a Zen Buddhist monk who regards wanting nothing as a type of power and strives for that. Such a pursuit does not appear threatening to society. Similarly, if AGIs pursued non-dangerous types of power, they might pose little threat to human society.

Indeed, arguments for existential threats typically follow two lines of reasoning, each assuming AGIs seek particular types of power: (i) that AGIs and humans will compete for the same resources, endangering humanity (Bostrom 2014; Ngo et al. 2023; Bales 2024; Dung 2024; Carlsmith 2023) and (ii) that by acquiring power, AGIs will become so capable of causing harm that if they have misaligned objectives, they could seriously harm humanity (Cotra 2021; Dung 2024; Carlsmith 2023).[3]

So, it is important to determine which types of power AGIs would pursue. One question arises: why would AGIs pursue the particular types of power emphasized in the existing literature rather than others? The underlying reason seems to be that these powers are useful in human society, and since AGIs will operate within that society, they too will find these powers useful and pursue them.

Although this idea may seem plausible, the following sections will cast doubt on it. The core intuition is that AGIs may understand and predict the world differently from humans, and thus may not identify the same types of power as suggested.

3 Avoiding anthropomorphism regarding AGIs' world models

This section begins by illustrating the role of AGIs' world models and then challenges the anthropomorphic assumption regarding these models.

[3] Some authors draw on both lines of argument to support the claim that AGIs may pose existential threats.

3.1 The instrumental convergence thesis and world models

Importantly, under the Instrumental Convergence Thesis, a subgoal only counts as instrumentally convergent if *AGIs themselves recognize it as such*. Otherwise, it does not follow that AGIs will pursue it. Put another way, if a subgoal is considered useful by humans but is not regarded as such by AGIs themselves, it cannot be an instrumentally convergent subgoal for AGIs.

This raises the question: what types of power would AGIs themselves regard as useful? Do these coincide with those emphasized in the existing arguments? Instrumental reasoning depends not only on AGIs' desires or final goals, but also on their *beliefs*, especially their *world models*.[4] To illustrate, AGIs' world models encode their understanding and predictions of world dynamics. These models, in turn, guide their actions.[5] Consider a simple illustration from a human world model. When playing baseball, a batter must quickly decide how to move in response to the opponent's actions and the ball's trajectory. This decision relies on the batter's understanding of baseball dynamics and on their prediction of the ball's future path. In this simple decision-making process, the batter's understanding of baseball dynamics constitutes a part of his world model and guides his actions.[6]

As many have suggested, existing and future AIs, especially AGIs, use world models to deliberate. For example, Dreamer explicitly implements a world model to predict future states and support planning (Hafner et al. 2019). Some have suggested that Sora exhibits early signs of world modeling through its ability to simulate physical outcomes (Cho et al. 2024). Moreover, some suggest that current large language models (LLMs) already encode latent world knowledge that enables them to make intuitive predictions about real-world dynamics (Manvi et al. 2023).[7] Likewise, robots utilize world models to represent and predict aspects of their environment, enabling action planning and task execution (Sakagami et al. 2023). Similarly, AGIs are expected

[4] We do not assume that AGIs literally possess beliefs; it suffices for our discussion that they act as if they do.

[5] The precise definition of a world model remains contested (Ding et al. 2024). In particular, some emphasize understanding (Ha and Schmidhuber 2018), others highlight the importance of prediction (LeCun 2022), and still others propose more specific operational definitions within domains such as robotics (Sakagami et al. 2023). However, this definitional debate does not substantially affect our argument here; readers may rely on the informal notion of a world model to proceed. We return to a more detailed discussion of world models in Sects. 3.2 and 6.2.

[6] This example comes from Ha and Schmidhuber (2018).

[7] For more on whether today's LLMs have beliefs, see Herrmann and Levinstein (2025).

to learn and use world models for planning toward a variety of goals (LeCun 2022).

If we misspecify AGIs' world models, systematic divergences can arise between human expectations and their behaviors. This leads to some important questions: what subgoals would AGIs identify *given their own world models*? Would they find the commonly discussed types of power instrumentally useful, or are such subgoals specific to human-like world models?

While previous work has emphasized the need to avoid anthropomorphism about AGIs' desires or goals, it has largely overlooked the aspect of beliefs, particularly world models. The upcoming section argues that we should also avoid anthropomorphism regarding AGIs' world models, and, once we do so, AGIs may not identify the types of power emphasized in the literature as useful subgoals.

One might suggest that presuming AGIs possess world models, or that they employ such models to guide their behavior, already reflects a form of anthropomorphism. While this may be a valid concern, two points justify this assumption for present purposes. First, as mentioned, many AI researchers explicitly argue that advanced AI systems will use world models in this way. Second, and more importantly, existing arguments linking the Instrumental Convergence Thesis to existential threats already presuppose that AGIs rely on some form of world models to reason about their instrumental goals. Turner et al. (2019), for example, frame their argument in terms of a Markov decision process, where probability functions encode AGIs' world models about how the environment changes, thereby constraining their options. Similarly, Gallow (2024) models AGIs' behavior within a sequential decision-making framework, where beliefs are represented probabilistically and used to calculate expected utility. Carlsmith (2023, p. 5), likewise, assumes that AGIs "make and execute plans, in pursuit of objectives, on the basis of models of the world". Our discussion proceeds on this basis.

However, these arguments often overlook a further anthropomorphic assumption: that AGIs' will represent world dynamics in ways similar to how humans do. In other words, the issue is not the existence or the role of such models, but the expectation that their structure and content will resemble our own. It is this specific kind of anthropomorphism that the present discussion aims to challenge 2023.

3.2 We should avoid anthropomorphism regarding AGIs' world models

Just as we have no reason to assume AGIs' final goals will mirror those of humans, we have no intuitive reason to assume AGIs' world models will resemble ours. It is important to note that, although AGIs are expected to be at least as capable as humans in many domains, it does not follow that AGIs will have the same world models as humans across those domains. To illustrate this point, consider LLMs. LLMs can perform many tasks at or beyond human level, such as answering questions, writing essays, or generating code (Bubeck et al. 2023). Yet the internal models that LLMs use for generating responses may be very different from the models used by humans (Hao et al. 2023). When humans answer questions or write essays, we rely on an internal model of language: we ensure our responses are grammatically correct, logically coherent, and appropriate for the audience. By contrast, LLMs produce text by predicting the most likely next words, using statistical patterns learned from massive training data.

Although the statistical patterns may sometimes cohere with human understanding of grammar, logic, or communication, they can also diverge significantly. One striking example is the case in which, when asked whether 9.11 is greater than 9.9, some LLMs have answered "yes". This is because in the contexts LLMs saw during training, "9.11" often comes after "9.9" (as in dates, where September 11 follows September 9) (Browne 2025). This illustrates how LLMs' world models (in this case, their models of numbers) can diverge from ours.

To be clear, evaluating the performance of AI systems is multidimensional. Even if current LLMs misunderstand numerical reasoning in such cases, they still outperform humans in many important text-related tasks and are thus considered highly capable. Importantly, their statistical understanding of the world, as mentioned above, in contexts, such as question answering or code writing, often yields results that surpass those of humans. Therefore, this kind of divergence does not imply that they are less capable (with respect to text generation). Indeed, given their remarkable capabilities, recent research has begun to explore the use of LLMs as backbones for constructing world models used for a wide range of decision-making tasks (see, e.g., Xing et al. 2025 for a recent proposal).[8]

In general, if one argues that sufficiently capable AGIs should have human-like world models, they are *effectively presupposing that only models resembling those of humans can yield good performance*. However, this presupposition is unjustified. On one hand, it is unclear whether humans actually understand the world well; it is possible that AGIs will form better yet different world models than humans. On the other hand, it is unclear whether there exists a single, universally correct understanding of the world, especially if "correct" is defined pragmatically as leading to good performance. As suggested by LLM applications, AGIs might

[8] See also Ding et al. (2024) for a recent survey of LLM-based world models.

develop different world models from those of humans while still performing at or above human levels.

The above considerations parallel the motivations for avoiding anthropomorphism regarding AGIs' desires or final goals: there is no obvious reason to assume that AGIs will have the same world models as humans, or that being as capable as humans implies having human-like world models.

We also have independent reasons to think that AGIs' world models may differ significantly from those of humans. The difference in world models may stem from differences in how AGIs and humans generate their world models. Humans generate their world models through a combination of embodied experience, perceptual learning, social interaction, cultural transmission, evolutionary pressures, etc. AGIs, by contrast, may generate their world models through entirely different pathways. For example, if we consider deep learning-based AGIs, their world models may be learned by optimizing over specific datasets, reflecting statistical regularities present in the training data. After training, AIs may identify different factors and correlations in the data than humans do.

We now offer a more detailed explanation of how AGIs might develop world models that diverge from ours, focusing on the context of model-based reinforcement learning (MBRL), under the standard assumption of a Markov Decision Process framework. This setting aligns with many existing discussions (Turner et al. 2019; Hafner et al. 2025). For expository clarity, let us treat a world model abstractly as a probabilistic function capturing state transitions.[9] The analysis below parallels our earlier discussion of LLMs while highlighting sources of divergence concerning world models.

Crucially, unlike traditional AIs, AGIs' world models cannot be hand-crafted in advance, especially as they operate in complex and open-ended environments. Instead, AGIs must learn a world model by extracting structure from their own *observations* and the *reward signals* they receive.[10] By contrast, humans, as mentioned above, construct world models through embodied experience and social learning, developing abstract and generalizable causal structures. The difference in these two processes may lead AGIs to learn different world models from those of humans.

To begin, consider the states themselves. As discussed earlier, each state encodes what the agent takes to be the relevant features of the environment. In the case of a human batter, these may include the ball's initial velocity, spin, release angle, etc. Likewise, AGIs must represent their states in terms of the features they deems relevant. However, due to differences in the learning process, AGIs may encode a different feature set from that of humans. As illustrated in Bobu et al. (2024), in the case of a robot tasked with cleaning dust from the floor, the reward is defined as "maximize dust collected." In such cases, the robot's learned state features may include only the amount of dust present and the amount removed, since these are directly tied to reward. A human, by contrast, would likely identify different features as salient, such as the net improvement in cleanliness, relative to the initial state.

Even when AGIs identify the same state features, the relationships they encode between states may still diverge significantly from those in human world models. This again stems from reward-driven optimization. Continuing with the dust-cleaning example, the robot may come to believe that dumping additional dust onto the floor increases the reward potential, as more dust can then be swept up. From the robot's perspective, the transition from "clean floor" to "dusty floor" and then "dust swept" forms an effective loop for maximizing cumulative reward. Yet such a transition structure clearly differs from world dynamics as understood by humans.

As many have noted, it is challenging to ensure that AIs acquire human-like world models. On one hand, we often lack the data necessary to ensure that AIs identify the same features and correlations that humans do. Perhaps, the real world is far too complex for current datasets to capture comprehensively. On the other hand, the black-box nature of modern deep learning systems makes it extremely difficult to monitor or verify the internal world models learned by AIs, let alone determine whether they cohere with human models (Ha and Schmidhuber 2018; Bobu et al. 2024). Consequently, AGIs may just learn fundamentally different world models from those of humans.

Given the above, we should avoid anthropomorphism regarding AGIs' world models. The upcoming sections argue that, once we set aside this assumption, it becomes unclear not only whether AGIs will pursue the emphasized types of power, but also which types of power AGIs would pursue at all.

[9] A full world model in this setting typically includes a reward model and an action set. The transition dynamics involve action–state interactions. Formally, let S be a non-empty finite set of states. Let A be a set of actions available to the agent. The goal is to learn a probabilistic transition function p that assigns to each state $s\prime$ in S a probability given the previous state s in S and an action a in A, i.e., $p(s\prime|s,a)$. Alternatively, this can be represented as a stochastic transition function T from $S \times A$ to $\Delta(S)$, where $\Delta(S)$ denotes the set of all probability distributions over S. For illustrative purposes, we adopt this simplified formulation here.

[10] Formally, we assume that each underlying state s probabilistically generates data, from which the agent receives a sequence of observations $o1, \ldots, oR$. The agent then constructs a task-relevant internal state representation sR by applying a learned mapping—typically via an encoder—from $o1, \ldots, oR$.

4 Re-evaluating the emphasized types of power

The existing arguments for existential catastrophe often implicitly assume that AGIs will develop human-like world models. Take, for example, the widely discussed idea that AGIs will seek to avoid shutdown (Turner et al. 2019; Carlsmith 2023). Building on this idea, some have proposed ways to design advanced AIs that either do not avoid shutdown or even actively seek it (Thornley 2024; Goldstein and Robinson 2024). However, why should we expect AGIs to avoid shutdown in the first place? The common explanation draws an analogy between shutdown for AGIs and death for humans. Turner et al. (2019, p. 10), for instance, write: "Intuitively, survival is power-seeking relative to dying, and so shutdown-avoidance is power-seeking behavior". Yet this analogy already assumes that AGIs conceptualize shutdown in the same way humans conceptualize death. In other words, it reflects a form of anthropomorphism regarding AGIs' world models: it assumes AGIs would understand "shutdown" the same way humans understand (AGIs') death.

If we drop that assumption, the analogy loses force. AGIs might not interpret shutdown—whether temporary or permanent—in the same terms as human mortality. Temporary shutdown, for instance, may not be perceived as death if the AGIs expects to be reactivated. Even if a shutdown is permanent, AGIs might not regard it as the end of their existence; what humans consider "permanent shutdown" (such as hardware destruction or software erasure) might not be regarded by AGIs as final. They might regard backups, copies, or updated versions as continuations of themselves, implying that the destruction of a single instance does not constitute death from the AGI's perspective.[11]

One might object that the existing arguments do allow some divergence in AGIs' world models. Indeed, they acknowledge that AGIs would not experience biological death, and instead treat "shutdown" as the functional equivalent for them. Yet even so, those arguments remain tied to human-like assumptions. Such reasoning often takes the form: "*If I were in AGIs' position, how would I reason about this situation?*" But what truly matters is not how humans would reason if placed in AGIs' shoes, but how AGIs themselves would reason, given their own perspectives and models of the world. The more relevant question, indeed, is: "*How would AGIs reason, given their own understanding and prediction of the situation?*" Regarding the shutdown example, what matters is how AGIs would interpret "death" within their own world models. That requires asking what factors AGIs would regard as relevant to their own termination, and how they would encode the relationships among those factors in their world models. However, these have been largely overlooked in the existing literature.

A similar anthropomorphic assumption appears in arguments regarding the disempowerment of humanity. Many suggest that AGIs will treat disempowering humanity as an instrumentally useful subgoal. Dung (2024, p. 10), for example, writes: "Disempowering humanity would be very useful for a wide range of goals, since it rules out a likely source of interference with its goals and provides access to enormous resources." But again, this presumes that AGIs would understand human disempowerment in this way. Why should we expect AGIs to draw the same connection between disempowering humans, reducing interference, and acquiring resources? This relies on a particular view of world dynamics that AGIs may not share.

This anthropomorphism goes beyond shutdown or human disempowerment. It underlies much of the current speculation about the types of power AGIs will pursue, such as self-preservation, goal preservation, cognitive enhancement, technological development, and resource acquisition. Whether AGIs will regard these as worth pursuing depends on how they understand them within their own world models. If AGIs' world models differ from ours, then what counts as power may shift accordingly. If we drop anthropomorphism regarding AGIs' world models, it is unclear whether AGIs would pursue any of these emphasized types of power.

Some may argue that subgoals such as self-preservation, goal preservation, cognitive capability enhancement, technological development, and resource acquisition are extremely coarse-grained and generic—so much so that they resemble the concept of "power" itself. From this perspective, such subgoals might be pursued by any sufficiently capable agent with instrumental reasoning, regardless of its underlying world model. On this view, identifying these subgoals does not obviously presuppose a human-like world model.

This view is plausible—so long as these terms are construed broadly and coarsely. However, it is important to note that in many current arguments linking the Instrumental Convergence Thesis to existential threats, such coarse-grained subgoals are anthropomorphically mapped onto more specific, fine-grained subgoals that are potentially harmful to human society. As mentioned earlier (cf. Section 2), a number of authors claim that AGIs will acquire money or compute as means of resource acquisition (Ngo et al. 2023; Carlsmith 2023), avoid shutdown to preserve themselves (Dung 2024), or manipulate humans to better achieve these subgoals (Ngo et al. 2023; Dung 2024; Carlsmith 2023). Once we abandon anthropomorphism regarding AGIs' world models, it becomes far less obvious that these

[11] We note that Turner et al. (2021) treat permanent shutdown as a kind of "terminal state," meaning that no further transitions occur beyond it. Our concern here is more specific: why should we expect AGIs themselves to represent hardware destruction or software erasure as a terminal state within their own world models?

coarse-grained subgoals will translate into the particular fine-grained subgoals currently emphasized. This is again illustrated by the examples of shutdown and disempowerment of humanity discussed above. AGIs might possess non-human world models and, therefore, conceptualize these subgoals differently, possibly rejecting certain fine-grained, human–salient strategies as instrumentally irrelevant or even counterproductive. As a result, the link between the Instrumental Convergence Thesis and existential threats loses much of its strength.

5 Deeper uncertainty and risk

Worse still, once we take AGIs' world models into account, the uncertainty becomes deeper than previously recognized. It is not only unclear whether AGIs will pursue the types of power emphasized in the existing literature, but also which types of power they might pursue at all. We tend to identify certain types of power as being instrumentally useful, based on our experience within human society. However, if we abandon anthropomorphism regarding AGIs' world models, it becomes plausible that AGIs may identify types of power that do not align with human-defined categories, or that we have not yet conceptualized.

Notably, such divergence need not stem from drastic differences between AGIs and human world models. As earlier illustrated, even subtle divergences—such as different interpretations of shutdown situations—could lead AGIs to identify different types of power. Crucially, some of these types might directly conflict with those emphasized by humans. In such cases, AGIs may prioritize entirely different types of power from humans.

Indeed, the uncertainty is deeper than currently recognized. As argued earlier, we cannot resolve it by projecting from human viewpoints and modestly adjusting expectations as if we were in AGIs' shoes. What matters is how AGIs themselves, given their potentially non-human world models, would reason about power. In addition, for those who suggest that AGIs might still recognize some familiar types of power even with non-human world models, it is important to explain why this would follow from their models—something the current literature rarely addresses.

Some might reply that the observed power-seeking behaviors in today's advanced AI systems suggest convergence on familiar types of power. For instance, some systems appear to resist shutdown, manipulate users, or subtly seek control over resources (Cotra 2021; Hadshar 2023; Ngo et al. 2023; Ji et al. 2024). Yet this observation demands caution, as these observations remain scattered, limited, and often speculative. There is no robust evidence of systematic *convergence* in current frontier systems.

More importantly, existing monitoring efforts remain framed within human-defined taxonomies of power, such as resisting shutdown or seeking resource control. However, if AGIs develop non-human-like world models, they may pursue types of power outside these established categories. Such behaviors could still pose serious risks. However, they may remain undetected, simply because they fall outside our current frameworks of power-seeking. This oversight can be dangerous, precisely because we are entirely unprepared.

Some may reasonably suggest that AGIs could seek power in ways that are ultimately beneficial to humans. For example, AGIs may develop world models quite unlike our own. Given such differences, they may identify novel strategies for acquiring power. For instance, AGIs might discover methods for improving technological efficiency or chip design that humans have overlooked, and use such breakthroughs as a means of resource acquisition. This is certainly possible. However, the point remains: precisely because their world models may diverge from ours, the ways in which AGIs seek power may turn out to be even more difficult to predict than the current literature assumes. In addition, this deeper uncertainty merits closer examination.

6 Aligning AGIs' world models

To address the risk, we argue that it is important to align AGIs' world models. We then turn to several open questions for future research.

6.1 The importance of world model alignment

As widely recognized, alignment has been proposed as a central approach to addressing key challenges in developing advanced AI systems and mitigating AI risks (Russell 2019). Indeed, mitigating harmful power-seeking behavior in AGIs is often framed as a subproblem of AI alignment, with such behavior understood as a specific form of misalignment (Ji et al. 2024; Carlsmith 2023).

Broadly speaking, the goal of AI alignment research is to ensure that AI systems *do what we want them to do*. On this general formulation, one might expect world model alignment to be implicitly included, since world models directly influence decision-making and behavior. Unfortunately, this is not the case; existing literature has overwhelmingly focused on *value alignment*. For example, many authors use the terms "AI alignment" and "value alignment" interchangeably (Russell 2019; Gabriel 2020; Hendrycks and Mazeika 2022), while Gabriel (2020, p. 422) explicitly defines "value" in this context as referring to "natural or non-natural facts about what is good or bad, and about what kinds of things ought to be promoted." As a result, alignment research has largely focused on specifying human values and

embedding them into AI systems (Kim et al. 2021; Riesen and Boespflug 2025).[12]

However, as illustrated above, world models play a crucial role in AGIs' decision-making and behavior. Neglecting world model alignment may, therefore, result in alignment failure.

First, it may lead to misleading assessments of alignment. A system might have correctly specified values but operate on a misaligned world model. Because its values appear aligned, it may pass conventional alignment evaluations yet still behave dangerously in open-ended or real-world contexts. For example, AGIs might have no desire to avoid shutdown, yet interpret shutdown-avoidance behaviors as cooperative actions.

Second, neglecting world model alignment may lead to ineffective or wasted training efforts. Current alignment efforts often focus heavily on specifying values and providing ethically curated data to guide value learning. However, if the real issue stems from flawed world models, such efforts alone are unlikely to ensure safe behavior. That is, without attention to align AGIs' world models, refining values alone may be insufficient to ensure safety.

By contrast, explicitly considering world model alignment may help us identify the true sources of alignment failures—whether misaligned values, misaligned world models, or both. This distinction could inform both data curation and the development of more targeted alignment techniques. For instance, if failures stem primarily from flawed world models, we should redirect resources toward correcting or refining AGIs' world models, rather than endlessly refining values.

6.2 Open questions

But how can we align AGIs' world models? We close by proposing several open questions to advance inquiry in this area.

We may define the problem of world model alignment as the question of *whether AI systems possess world models that we want them to have*. In contrast to value alignment, which concerns what values an AI system should have, world model alignment raises a parallel but distinct question: what types of models of the world do we want AGIs to adopt? In other words, what is our goal for aligning AGIs' world models?

Particularly, human values typically favor beliefs that are true, responsive to evidence, and conducive to desirable practical outcomes. However, should AGIs adopt world models with these same properties? Are they always desirable in the context of AGI safety and alignment? The answer is far from obvious. Consider the case of permanent shutdown. Whether or not it is strictly true that shutdown constitutes death for AGIs—assuming such a claim can even bear a truth value—we might nonetheless prefer AGIs to believe that it does not. In this sense, truth may not always be the over-riding criterion in AGIs' world models: at a minimum, AGIs should adopt world models that are unlikely to lead to harmful power-seeking behavior—or that explicitly rule it out.

To make the above questions more concrete, we now turn to how current architectures construct world models. In Sect. 3.2, we introduced two illustrative examples—LLM-based and MBRL-based world models—to establish an initial conceptual understanding of world models. More systematically, recent surveys suggest that existing approaches can be broadly categorized into two methodological paradigms: those that prioritize *understanding the world* and those that emphasize *predicting the future* (Ding et al. 2024; Xing et al. 2025).

Approaches focused on understanding aim to learn accurate and generalizable representations of world dynamics. They typically adopt an encoder–decoder architecture and are trained to minimize *reconstruction error*, often regularized by Kullback–Leibler (KL) divergence. A classic example is Ha and Schmidhuber's (2018) world model architecture, which uses a variational autoencoder (VAE) for image reconstruction and a recurrent mixture density network to model latent state transitions.

In contrast, predictive paradigms emphasize learning latent-space transitions directly, without reconstructing the observation. These models usually employ encoder-only architectures—the observation reconstruction step is bypassed—and are trained to minimize *prediction error* between predicted and target embeddings. The Joint Embedding Predictive Architecture (JEPA) proposed by LeCun (2022), for instance, explicitly avoids reconstructing raw observations. Instead, it maps current contextual embeddings to predicted future embeddings, minimizing a divergence between internal representations.

Regarding these two paradigms, we have several open questions, particularly concerning *training objectives* and *architectural choices*. First, while they differ methodologically, both prioritize some notion of accuracy—whether minimizing reconstruction error in encoder–decoder systems or representational prediction error in encoder-only architectures. However, in the context of world model alignment and safety, it is worth questioning whether accuracy, as conventionally defined, should always be our guiding objective. As

[12] Varshney et al. (2025) argue that AI alignment should consider not only an AI's values but also its knowledge. Their focus, however, lies primarily in enabling AIs to competently perform tasks, rather than in addressing the potential risks of deliberately harmful behavior stemming from misaligned world models. In contrast, our approach focuses specifically on this latter concern.

illustrated by the shutdown example, it may sometimes be safer for AGIs to maintain functionally benign but factually inaccurate beliefs. Put another way, we might ask: should we aim to construct AGIs with normatively desirable world models, even at the expense of strict empirical accuracy? And if so, how should such desirable models be specified, measured, or learned?

Recent work on representational alignment (Sucholutsky et al. 2023; Bobu et al. 2024) has focused on increasing representational similarity between systems—such as between AI agents and humans—and has introduced tools to measure and improve such similarity. Although these methods were not originally designed with safety issues, they offer valuable starting points. Rather than enforcing representational similarity for its own sake, we might repurpose these tools: they could help estimate and enhance the alignment between AGIs' learned world models and a human-endorsed or safety-prioritized reference model.

Second, we may ask: which types of architectures are more conducive to world model alignment? Architectural choices bear directly on the tractability of alignment. Xing et al. (2025) argue that encoder–decoder frameworks, by enabling reconstruction of observable data, allow for stronger and more interpretable supervision. Such models make it easier to assess whether the learned internal representations correspond to meaningful and human-relevant aspects of the world. In contrast, encoder-only models operate solely in latent space and focus on task-relevant features. This may make it more difficult to verify whether the learned representations are meaningful or robust. In our context, this also raises the question of whether architectures that support observation-grounded supervision are better suited to align with human-endorsed or safety-prioritized reference models.

Furthermore, architectural considerations also arise in response to the changing world. Many components of the world are dynamic, and we may want AGIs' world models to remain adaptively aligned over time. In particular, world models must capture evolving social and normative structures, such as social interactions, legal regulations, and others' mental and emotional states (Li et al. 2023; Taniguchi et al. 2023; Ding et al. 2024). These structures vary across cultures and evolve over time, requiring world models to flexibly adapt. For example, urban environments (e.g., pedestrian behaviors) are constantly evolving, and urban world models must be able to track and integrate such changes (Zhang et al. 2025). These challenges raise important questions: how can AGIs' world models be made dynamically aligned? Which types of architectures best support this form of continual adaptation?

To address these questions, we may draw insights from psychology and cognitive science—two fields that have long inspired developments in AI. For instance, the architecture proposed by Ha and Schmidhuber (2018) was inspired by the human cognitive system. It incorporates modular components for vision, memory, and control, mirroring the structure of human mental models described in psychology (Gentner and Stevens 2014). Similarly, although LeCun's (2022) JEPA was motivated by efficiency in representation learning (e.g., avoiding unnecessary details), its emphasis on minimizing predictive errors in latent space shares conceptual roots with the Free Energy Principle and predictive coding frameworks originating in cognitive science (Friston et al. 2021; Taniguchi et al. 2023). Moreover, the distinction between fast reactive modules (mode 1) and slower predictive modules (mode 2) in JEPA draws clear inspiration from dual-process theories in psychology (Kahneman 2011).

Recent work has begun exploring these connections by designing architectures inspired by human cognitive development. For example, recent work in cognitive and developmental robotics explores the integration of predictive coding into world model architectures, with the aim of enabling embodied agents to continuously update internal representations through interaction with changing environments (Tani and White 2022; Taniguchi et al. 2023). In the context of value alignment, to address the problem of changing and conflicting human values, White (2024) proposes aligning AI with invariant moral ideals by designing AI agents whose internal motivational structures reflect the hierarchical development of human moral cognition. This design supports stable, lifelong value orientations. Liu et al. (2025) propose a modular, brain-inspired framework for building foundation agents that integrate LLMs with components for perception, memory, emotion, and planning. These agents are designed to support continual learning and self-evolution in complex, dynamic environments.

7 Conclusion

This paper critically examines the anthropomorphic assumptions underlying current arguments that link the Instrumental Convergence Thesis to existential threats. We argue that once anthropomorphism about AGIs' world models is dropped, it becomes unclear not only whether AGIs will pursue the types of power emphasized in the existing literature, but also which types of power they might pursue at all. This analysis challenges the strength of prevailing arguments and highlights an overlooked risk: AGIs with non-human world models may identify unfamiliar—and potentially dangerous—forms of power that fall outside current taxonomies. To address this risk, we argue that world model alignment—a largely neglected dimension compared to value alignment—is crucial for AI safety. We conclude by outlining several open questions to guide future research on aligning AGIs' world models.

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Declarations

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